



Tesla smokes Lamborghini

A recent drag race video posted on YouTube shows a Tesla Model X P100D beating a \$530,000 Lamborghini Aventador. <http://dgit.in/XLambo>



Deepmind vs Starcraft

Google's Deepmind and Blizzard just released tools to let AI researchers create bots capable of playing Starcraft. <http://dgit.in/StarcraftAI>

Bijayan Venus Mission

wants to terraform Earth's twin planet

An orbiter will spray extremophiles into the Venusian atmosphere, attempting to initiate terraforming on the planet

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erraforming is the process by which inhabitable planets can be made favourable for living on by

creatures of Earth. The process involves deliberately changing a planet's ecology and atmosphere, which goes on to affect various factors such as temperature, pressure, etc. The goal is to transform the conditions into something similar on Earth or suitable for earthlings. This has been a dream for decades, right since well-known astronomer Carl Sagan had proposed the same for Venus, followed by Mars. With a similar goal in mind, the Indian Astrobiology Research Centre (IARC) is working on a terraforming mission of Venus. Based in Mumbai and Surat, IARC's primary research is on Panspermia, or the hypothesis that life exists throughout the universe and is transported across asteroids, comets and planetoids. Bijayan Venus Mission wants to be the first project to initiate

the terraforming of the second planet from the Sun. The name "Bijayan" is derived from two Sanskrit words, where "bij" means seed and "ayan" means journey. Essentially, together they mean the journey of a seed. This "seed" will be a set of carefully selected microorganisms that will be injected into the planet's atmosphere to trigger the terraforming of Venus.

THE BIJAYAN VENUS MISSION

Stated to be "primarily a demonstration of intent", the main details of the mission have been listed on their website and only a few things have been finalised until now. The primary mission challenge is an orbital transfer of the spacecraft to Venus. It will rely on a flyby for orbital insertion of the payload into the atmosphere, hence it's being considered as an orbiter mission for now. The nano-spacecraft will be built in Mumbai and Surat, with a lower limit of 3 kg and an upper limit of 10 kg. The final mass of the entire spacecraft hasn't

been finalised as it depends on the configuration of the payload. India is already known to have successfully carried out cost-effective outer space missions in the past. However, those were official ISRO missions whereas the Bijayan Venus mission is being planned solely by the IARC in collaboration with other institutes and organisations. The team is trying to keep this mission economical and only sticking to one primary goal of successfully delivering the payload into the Venusian atmosphere. Bijayan Venus is the first outer space mission planned by IARC and probably the first in India. The conceptualisation and initial blueprints along with the collaborations will be self-funded under the Indian Astrobiology Research Funds (IARF). The primary projected cost allocated is around USD 25,000. Once this stage has been passed, they will be looking for other sources of funding including individuals who are interested